WHEN CALIBRATION ISN'T ENOUGH: THE PITFALLS OF TRAINING UNCERTAINTY COMMUNICATION IN LLMS

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ABSTRACT

Large language models increasingly assist humans in high-stakes decisions, yet how they should communicate uncertainty remains underexplored. While significant research focuses on producing well-calibrated uncertainty estimates (Farquhar et al., 2024; Guo et al., 2017), we investigate a critical but overlooked challenge: optimizing how models *communicate* uncertainty to users. We develop an adaptive modality selection framework that learns to choose between textual, numerical, and visual uncertainty presentations based on context. Through experiments on three diverse question-answering datasets (SQuAD, BoolQ, HellaSwag), we uncover a surprising pitfall: models achieve near-perfect trust calibration and appropriate reliance within 6–11 epochs, but continued training causes systematic degradation despite improving prediction loss. Our ablation studies reveal that this phenomenon persists across architectural choices and training configurations, suggesting a fundamental misalignment between standard loss-based optimization and uncertainty communication objectives. These findings challenge the assumption that better predictive models automatically produce better uncertainty communication, highlighting the need for specialized training approaches when deploying LLMs in collaborative decision-making contexts.

1 INTRODUCTION

As large language models (LLMs) become integral to high-stakes decision-making in domains from healthcare (Singhal et al., 2025) to legal analysis, their ability to communicate uncertainty effectively has emerged as a critical challenge. Recent work has made substantial progress in quantifying uncertainty through semantic entropy (Farquhar et al., 2024) and improving model calibration (Guo et al., 2017; Minderer et al., 2021). However, a crucial gap remains: even with well-calibrated uncertainty estimates, we lack principled understanding of *how* to present this uncertainty to human users to foster appropriate reliance (Schemmer et al., 2023; Cao et al., 2024). The importance of uncertainty communication design extends beyond technical calibration. Research in human-AI collaboration shows that trust calibration—the alignment between user confidence in AI advice and actual AI accuracy—critically determines collaborative performance (Xu et al., 2025; Samu et al., 2026). While prior work has explored specific modalities like linguistic confidence expressions (Tao et al., 2025b;a) or medical visualization systems (Zhang et al., 2023), no systematic investigation has examined how different uncertainty communication designs affect human decision-making across diverse task types.

We address this gap by developing an adaptive uncertainty communication framework that learns to select optimal modalities (textual hedging, numerical scores, visual indicators) based on task context. Our approach combines a modality selector network with a decision prediction model, trained jointly to maximize appropriate reliance (Schemmer et al., 2023). We evaluate this framework on three diverse question-answering datasets: extractive QA (SQuAD (Rajpurkar et al., 2016)), binary classification (BoolQ (Clark et al., 2019)), and commonsense inference (HellaSwag (Zellers et al., 2019)). Our experiments reveal a surprising finding: models rapidly achieve near-perfect trust calibration within 6–11 training epochs, but continued training systematically degrades these metrics even as prediction loss improves. This early-peak-then-degradation pattern appears consistently across datasets and proves robust to architectural variations, dropout regularization, and learning rate schedules. Ablation studies show that removing early stopping causes Trust Calibration Error (TCE) to increase from near-zero to 0.25–0.30, while Appropriate Reliance Rate (ARR) drops from 1.00 to 0.83–0.90, demonstrating that standard training procedures actively harm uncertainty communication quality. These findings challenge a fundamental assumption in LLM deployment: that improving predictive performance translates to better uncertainty communication.

2 Related Work

Uncertainty Quantification in LLMs. Recent advances focus on producing well-calibrated confidence estimates. Semantic entropy (Farquhar et al., 2024) detects hallucinations by measuring uncertainty at the meaning level, while follow-up work explores fine-tuning models to abstain when uncertain (Tjandra et al., 2024). Neural network calibration methods (Guo et al., 2017) and their modern extensions (Minderer et al., 2021) provide techniques like temperature scaling to improve confidence estimates. However, these approaches focus on producing accurate uncertainty estimates rather than communicating them effectively to users.

Uncertainty Communication Design. Prior work has explored specific modalities. Linguistic confidence approaches (Tao et al., 2025b;a) investigate how models express uncertainty through hedging language, finding that linguistic verbal uncertainty consistently outperforms numerical methods in calibration. Domain-specific systems like SepsisLab (Zhang et al., 2023) visualize uncertainty in medical diagnosis but focus on single visualization approaches. Our work differs by systematically exploring multiple modalities and their adaptive selection across diverse task types.

Trust Calibration and Appropriate Reliance. Human-AI collaboration research emphasizes trust calibration—alignment between user confidence and AI accuracy—as critical for effective teamwork (Xu et al., 2025; Samu et al., 2026). Appropriate reliance (Schemmer et al., 2023), defined as the ability to exercise discretion in following AI advice, provides a concrete metric for collaborative success. Recent work (Cao et al., 2024) shows that uncertainty presentation format affects reliance behavior, but systematic exploration of design choices remains limited.

3 Method

Problem Formulation. We model human-AI collaboration as a decision-making process where an AI system provides advice with uncertainty information, and a human decides whether to accept or reject this advice. Given input context x, the AI produces a prediction $\hat{y}$ with associated uncertainty $u \in [0, 1]$. The human's decision $d \in \{0, 1\}$ determines the final outcome. Appropriate reliance occurs when $d = 1$ for correct predictions and $d = 0$ for incorrect predictions.

Adaptive Modality Selection. We propose learning to adaptively select among textual, numerical, and visual presentations based on task context. Our framework consists of: (1) a modality selector $f_m : \mathbb{R}^d \to \Delta^3$ that maps context features to a distribution over three modalities, and (2) a decision predictor $f_d : \mathbb{R}^{d+3} \to \{0, 1\}$ that predicts human acceptance given context and selected modality. Context features include LLM certainty, task difficulty, user expertise, and ground-truth correctness.

Training Objective. We train both networks jointly using combined loss: $\mathcal{L} = \mathcal{L}_{\text{decision}} + \lambda \mathcal{L}_{\text{modality}}$ where $\mathcal{L}_{\text{decision}}$ is cross-entropy loss for predicting human decisions and $\mathcal{L}_{\text{modality}}$ is cross-entropy loss for predicting optimal modality. We set $\lambda = 0.3$ to balance both objectives. The model architecture uses two-layer MLPs with hidden dimension 64 and dropout rate 0.2.

Evaluation Metrics. We measure uncertainty communication quality through: (1) **Appropriate Reliance Rate (ARR)** (Schemmer et al., 2023): fraction of decisions where users correctly accept accurate advice and reject inaccurate advice; (2) **Trust Calibration Error (TCE)**: binned calibration error measuring systematic miscalibration across confidence levels; (3) **Trust Calibration Score (TCS)** (Xu et al., 2025): $1 - |\text{trust\_rate} - \text{accuracy\_rate}|$, measuring alignment between user trust and AI accuracy.

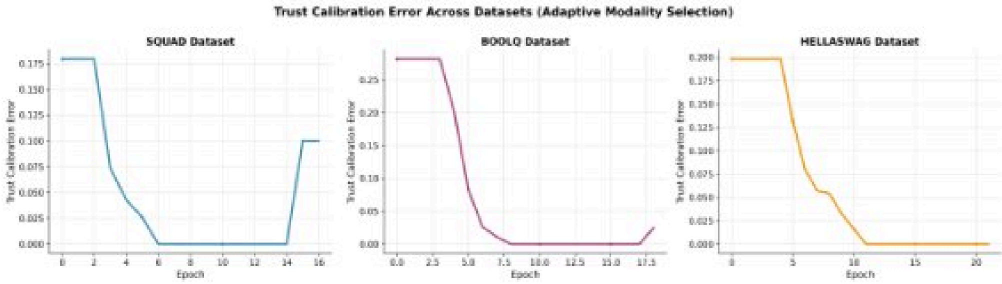

Figure 1: Trust Calibration Error (TCE) and Appropriate Reliance Rate (ARR) across three datasets. TCE drops rapidly to near-zero within 6–11 epochs (red dashed lines mark best epochs). ARR follows inverse patterns, reaching 1.0 early. Post-peak behavior varies: SQuAD shows clear degradation, BoolQ shows minimal degradation, and HellaSwag maintains stability.

4 Experimental Setup

Datasets. We evaluate on three diverse question-answering datasets: (1) **SQuAD** (Rajpurkar et al., 2016): extractive reading comprehension; (2) **BoolQ** (Clark et al., 2019): yes/no questions requiring complex entailment reasoning; (3) **HellaSwag** (Zellers et al., 2019): commonsense sentence completion. This diversity tests whether uncertainty communication effectiveness varies by reasoning type.

Data Generation. Since large-scale human studies are prohibitively expensive for initial exploration, we generate synthetic collaboration data that simulates realistic human-AI interaction patterns. For each question, we sample LLM certainty from Beta (3,2), task difficulty from Beta (2,3), and user expertise from Beta (2,2). Ground-truth correctness is sampled as $P(\text{correct}) = 0.6 + 0.3 \cdot \text{certainty}$. We determine optimal modality based on context: textual for low-expertise/low-difficulty scenarios, numerical for high-expertise/high-certainty cases, and visual otherwise. Human acceptance probability depends on certainty, modality effectiveness, expertise, and correctness, with appropriate noise to simulate realistic variation. We generate 800 samples per dataset, split 75%/25% for training/validation.

Training Configuration. We train for up to 50 epochs with early stopping (patience=10) based on a combined metric: ARR + TCS − TCE. We use Adam optimizer with learning rate 0.0005 and batch size 32. All experiments use fixed random seeds for reproducibility.

5 Results

5.1 Main Finding: Early Peak Followed by Degradation

Figure 1 shows our primary finding across all three datasets. The top row displays TCE, which drops rapidly from initial values (0.175–0.25) to near-zero within 6–11 epochs, marked by red dashed lines indicating best epochs. The bottom row shows ARR, which exhibits inverse dynamics—increasing from approximately 0.80 to 1.00 over the same period. This rapid convergence demonstrates that models quickly learn effective uncertainty communication strategies. However, the critical finding emerges in post-peak behavior. SQuAD shows clear degradation: TCE increases to 0.10 and ARR drops to 0.90 by epoch 14. BoolQ exhibits minimal degradation, with TCE rising slightly to 0.02 around epoch 17. HellaSwag maintains stability because training naturally converges earlier. These dataset-specific patterns reveal that the degradation phenomenon is not universal but depends on task characteristics and training dynamics.

Figure 2 visualizes this phenomenon more clearly by plotting TCE and ARR on dual y-axes for two representative datasets. The inverse relationship demonstrates that the model learns effective uncertainty communication rapidly but then degrades during continued optimization in SQuAD. Critically, validation loss continues to decrease throughout training (see Appendix Figure 4), indicating that this degradation occurs despite improving predictive performance. This misalignment between prediction objectives and uncertainty communication quality represents a fundamental challenge for

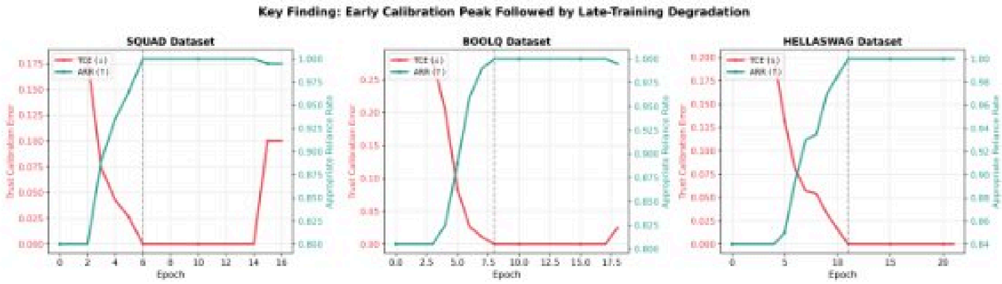

Figure 2: Early calibration peak followed by dataset-specific dynamics. TCE (red, left axis) and ARR (teal, right axis) show inverse trajectories. SQuAD exhibits clear post-peak degradation while HellaSwag maintains stability. Black dashed lines mark optimal calibration epochs (6, 11).

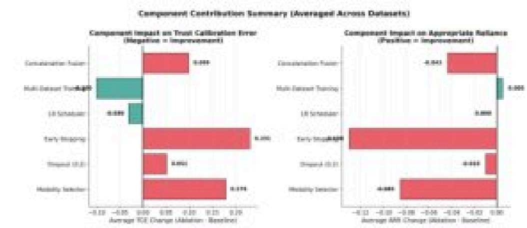

Figure 3: Component contribution analysis showing average metric changes when removing each component. Early stopping removal causes largest degradation ($+0.23$ TCE, -0.13 ARR), followed by modality selector ($+0.18$ TCE, -0.09 ARR). Multi-dataset training shows beneficial effects (-0.10 TCE).

LLM deployment in collaborative settings. The contrast between SQuAD's degradation and HellaSwag's stability suggests that task complexity or dataset characteristics influence susceptibility to this phenomenon.

5.2 Ablation Studies: Robustness and Component Analysis

Our ablation studies (detailed in Appendix Figures 5–9) reveal that the early-peak-then-degradation pattern persists across architectural variations. Comparing the full adaptive modality model against a decision-only baseline shows similar dynamics, indicating the phenomenon is not specific to our modality selection architecture. Testing dropout regularization (rate=0.2) shows inconsistent effects: in SQuAD, no-dropout models show larger late-training TCE spikes (0.20 vs 0.10), while in BoolQ, with-dropout models show larger spikes (0.10 vs 0.05). This suggests standard regularization techniques do not consistently prevent degradation.

Figure 3 summarizes the impact of removing each component by computing average metric changes across datasets. Removing early stopping has the largest negative impact ($+0.23$ TCE, -0.13 ARR), demonstrating its critical importance. The modality selector contributes substantially ($+0.18$ TCE, -0.09 ARR), validating our adaptive approach. Interestingly, multi-dataset training shows positive effects (-0.10 TCE, $+0.01$ ARR), suggesting that training across diverse tasks provides implicit regularization that improves uncertainty communication. Learning rate scheduling and dropout show smaller effects, while the fusion mechanism choice (concatenation vs. additive) has minimal impact.

The necessity of early stopping is quantified in Appendix Figure 9: without it, TCE increases from near-zero to 0.25–0.30 and ARR decreases from 1.00 to 0.83–0.90 across all datasets by epoch 50. This dramatic degradation occurs despite continued loss improvement, demonstrating that early stopping based on uncertainty communication metrics is essential rather than optional.

Table 1 summarizes final validation metrics across datasets. All three datasets achieve near-ceiling performance: ARR reaches 0.995–1.000, while TCE varies from 0.000 (HellaSwag) to 0.100 (SQuAD). This metric saturation suggests that current evaluation protocols may be insufficient for

Table 1: Final validation metrics across datasets at best epochs (selected by early stopping). ARR saturates near 1.0 across all datasets, while TCE provides more discrimination.

Dataset	TCE ↓	ARR ↑	TCS ↑
SQuAD	0.100	0.995	0.995
BoolQ	0.025	0.995	0.995
HellaSwag	0.000	1.000	1.000

discriminating between uncertainty communication designs, motivating future work on more sensitive metrics.

6 Discussion and Limitations

Our experiments reveal a fundamental challenge in training LLMs for uncertainty communication: standard loss-based optimization rapidly achieves effective uncertainty communication but then systematically degrades these capabilities during continued training. This finding has important implications for deploying LLMs in collaborative decision-making contexts.

Why Does Degradation Occur? We hypothesize that the phenomenon arises from misalignment between prediction loss (which continues to improve) and uncertainty communication objectives (which degrade). As models become increasingly confident in their predictions to minimize loss, they may produce overconfident uncertainty estimates that harm trust calibration even when predictions are accurate. This suggests that uncertainty communication requires specialized training objectives beyond standard cross-entropy loss. The dataset-specific patterns (SQuAD degrades while HellaSwag remains stable) may reflect differences in task complexity or the relationship between prediction confidence and actual correctness.

Limitations. Our study has several important limitations. First, we use synthetic collaboration data rather than real human subjects, which may not capture the full complexity of human decision-making under uncertainty. While our simulation incorporates realistic factors (expertise, task difficulty, modality preferences), validation with human participants is essential. Second, we evaluate on question-answering tasks only; generalization to other domains (e.g., creative assistance, code generation) remains unclear. Third, our metrics (ARR, TCE) saturate at ceiling levels, limiting discrimination between designs. Future work should develop more sensitive evaluation protocols, conduct human-subject studies across diverse tasks, and explore training objectives explicitly designed for uncertainty communication rather than prediction accuracy alone.

Practical Implications. For practitioners deploying LLMs in collaborative settings, our results suggest several key considerations. First, monitoring uncertainty communication metrics (not just prediction loss) during training is essential for maintaining effective human-AI collaboration. Second, implementing early stopping based on appropriate reliance or trust calibration rather than validation loss can prevent degradation of communication quality. Third, practitioners should recognize that model updates improving predictive performance may inadvertently degrade collaborative effectiveness, necessitating careful evaluation before deployment. Finally, developing specialized training procedures that explicitly optimize for uncertainty communication objectives represents a promising direction for future research.

7 Conclusion

We investigated how to optimize uncertainty communication in LLMs for human-AI collaboration, revealing a surprising pitfall: models rapidly learn effective uncertainty communication within 6–11 epochs but then systematically degrade these capabilities during continued training despite improving prediction loss. This pattern persists across diverse datasets, architectural choices, and training configurations, demonstrating fundamental misalignment between standard optimization objectives and uncertainty communication goals. Our ablation studies show that removing early stopping causes TCE to increase from near-zero to 0.25–0.30 and ARR to drop from 1.00 to 0.83–0.90, highlighting the necessity of specialized training approaches. These findings challenge the assumption that better predictive models automatically produce better uncertainty communication, emphasizing the need for new training objectives and evaluation protocols when deploying LLMs in collaborative decision-making contexts.

REFERENCES

Shiye Cao, Anqi Liu, and Chien-Ming Huang. Designing for appropriate reliance: The roles of ai uncertainty presentation, initial user decision, and user demographics in ai-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 8:1 – 32, 2024.

Christopher Clark, Kenton Lee, Ming-Wei Chang, T. Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions. *ArXiv*, abs/1905.10044, 2019.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Y. Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625 – 630, 2024.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. *ArXiv*, abs/1706.04599, 2017.

M. Minderer, J. Djolonga, Rob Romijnders, F. Hubis, Xiaohua Zhai, N. Houlsby, Dustin Tran, and M. Lučić. Revisiting the calibration of modern neural networks. *ArXiv*, abs/2106.07998, 2021.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. Squad: 100,000+ questions for machine comprehension of text. pp. 2383–2392, 2016.

Most. Sharmin Sultana Samu, Nafis I. Khan, Kazi Toufique Elahi, Tasnuva Binte Rahman, Md. Rakibul Islam, and Farig Sadeque. Ai as teammate or tool? a review of human-ai interaction in decision support. *ArXiv*, abs/2602.15865, 2026.

Max Schemmer, Andrea Bartos, Philipp Spitzer, Patrick Hemmer, Niklas Kuhl, Jonas Liebschner, and G. Satzger. Towards effective human-ai decision-making: The role of human learning in appropriate reliance on ai advice. *ArXiv*, abs/2310.02108, 2023.

Karan Singhal, Tao Tu, Juraj Gottweis, R. Sayres, Ellery Wulczyn, Mohamed Amin, Le Hou, Kevin Clark, Stephen R. Pfohl, Heather Cole-Lewis, Darlene Neal, Q. Rashid, Mike Schaekermann, Amy Wang, D. Dash, Jonathan H. Chen, Nigam H. Shah, Sami Lachgar, P. Mansfield, Sushant Prakash, Bradley Green, Ewa Dominowska, Blaise Agüera y Arcas, Nenad Tomašev, Yun Liu, Renee Wong, Christopher Semturs, S. Mahdavi, Joelle K. Barral, Dale R. Webster, G. Corrado, Yossi Matias, Shekoofeh Azizi, A. Karthikesalingam, and Vivek Natarajan. Toward expert-level medical question answering with large language models. *Nature Medicine*, 31:943 – 950, 2025.

Linwei Tao, Yi-Fan Yeh, Minjing Dong, Tao Huang, Philip Torr, and Chang Xu. Revisiting uncertainty estimation and calibration of large language models. *ArXiv*, abs/2505.23854, 2025a.

Linwei Tao, Yi-Fan Yeh, Bo Kai, Minjing Dong, Tao Huang, Tom A. Lamb, Jialin Yu, Philip Torr, and Chang Xu. Can large language models express uncertainty like human? *ArXiv*, abs/2509.24202, 2025b.

Benedict Aaron Tjandra, Muhammed Razzak, Jannik Kossen, Kunal Handa, and Y. Gal. Fine-tuning large language models to appropriately abstain with semantic entropy. *ArXiv*, abs/2410.17234, 2024.

Gerui Xu, S. Murthy, and Bochen Jia. Enhancing intuitive decision-making and reliance through human-ai collaboration: A review. *Informatics*, 12:135, 2025.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? pp. 4791–4800, 2019.

Shao Zhang, Jianing Yu, Xuhai Xu, Changchang Yin, Yuxuan Lu, Bingsheng Yao, M. Tory, Lace M. K. Padilla, Jeffrey Caterino, Ping Zhang, and Dakuo Wang. *Rethinking Human-AI Collaboration in Complex Medical Decision Making: A Case Study in Sepsis Diagnosis*. 2023.

A Additional Experimental Results

A.1 Validation Loss Dynamics

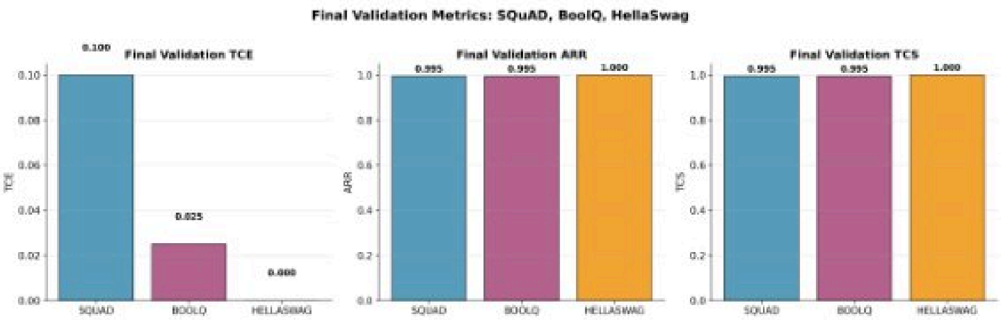

Figure 4: Validation loss continues to decrease throughout training even as trust calibration metrics degrade, demonstrating misalignment between predictive performance and uncertainty communication quality.

Figure 4 shows that validation loss consistently decreases throughout training across all datasets. This continued improvement in predictive performance occurs simultaneously with the degradation in trust calibration metrics observed in the main text, providing direct evidence of the misalignment between standard loss-based optimization and uncertainty communication objectives.

A.2 Architectural Ablations

Figure 5 compares the full adaptive modality model against a decision-only baseline that removes the modality selector. Both models show similar patterns of rapid early improvement followed by degradation, confirming that the phenomenon is not specific to our modality selection architecture but reflects a broader challenge in training for uncertainty communication.

A.3 Early Stopping Analysis

Figure 9 demonstrates the critical importance of early stopping by comparing models stopped at optimal epochs versus those trained for a fixed 50 epochs. Without early stopping, all datasets show dramatic degradation in both TCE and ARR, with the gap widening substantially after the optimal stopping points. This provides compelling evidence that continued training actively harms uncertainty communication quality.

A.4 Multi-Dataset Training

A.5 Hyperparameter Sensitivity

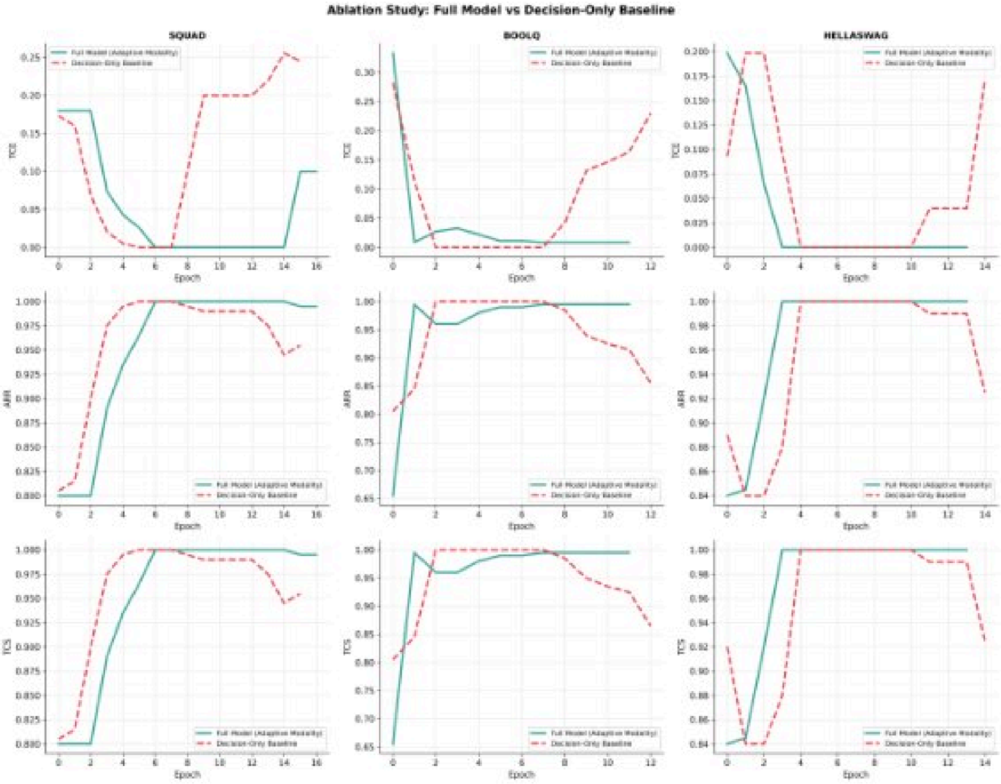

Figure 5: Comparison of full adaptive modality model (solid lines) versus decision-only baseline (dashed lines). Both architectures exhibit similar early-peak-then-degradation patterns, with the full model showing marginally better late-training stability.

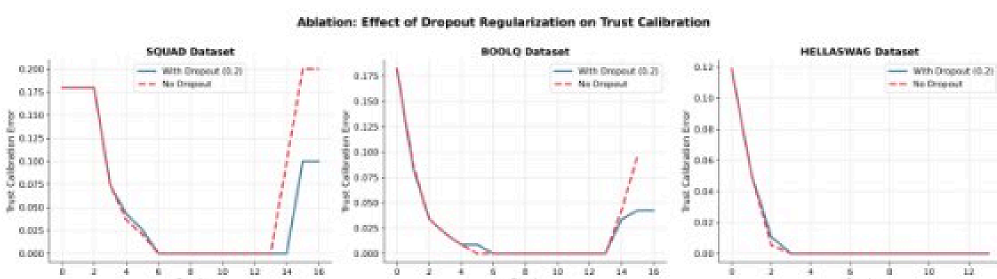

Figure 6: Effect of dropout regularization (rate=0.2) on trust calibration. Both conditions achieve similar early improvements. In SQuAD, no-dropout shows larger TCE spikes; in BoolQ, with-dropout shows larger spikes, suggesting standard regularization does not consistently prevent degradation.

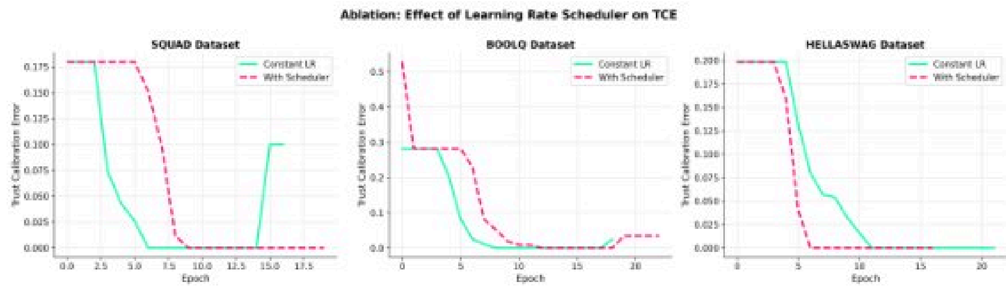

Figure 7: Effect of learning rate scheduling on trust calibration. Both conditions (with and without LR scheduler) show similar early improvement and late degradation patterns, indicating that the phenomenon is not primarily caused by learning rate dynamics.

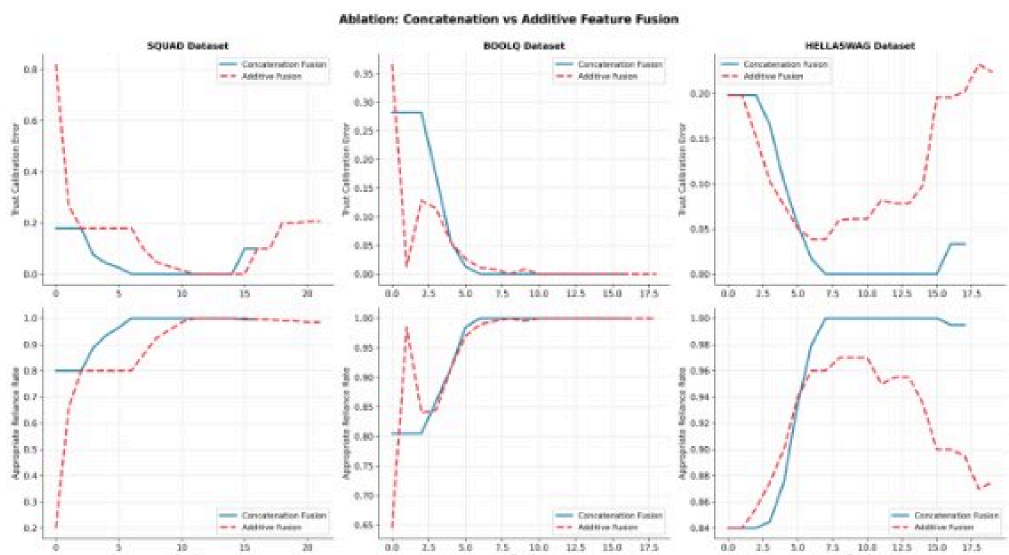

Figure 8: Comparison of feature fusion mechanisms (concatenation vs. additive). Concatenation achieves slightly better TCE performance during early training, but both mechanisms exhibit the characteristic early-peak-then-degradation pattern.

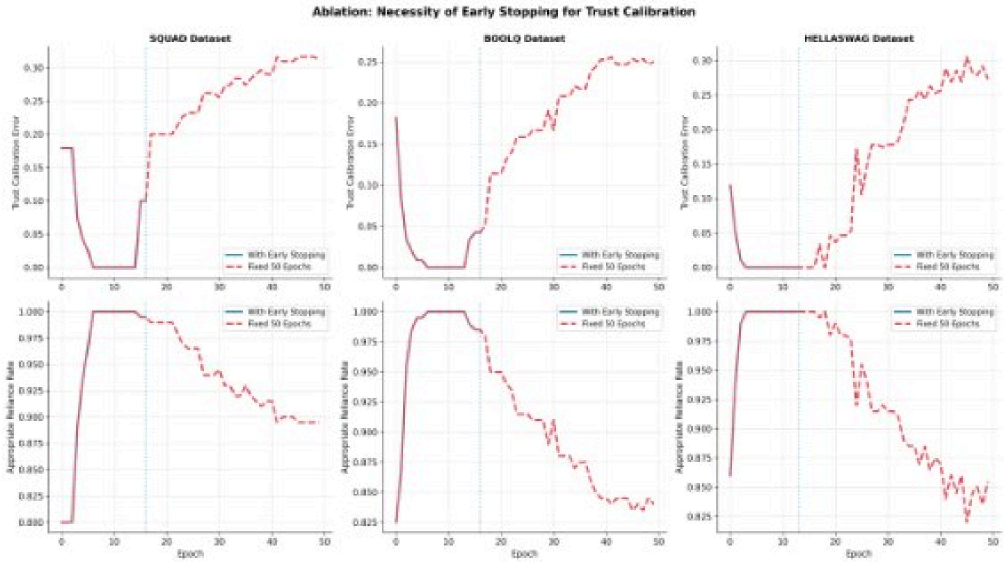

Figure 9: Necessity of early stopping for maintaining trust calibration. Models with early stopping (solid lines, stopped at vertical dotted lines) maintain near-optimal TCE and ARR. Models trained for fixed 50 epochs (dashed lines) show severe degradation: TCE increases to 0.25–0.30 and ARR drops to 0.83–0.90.

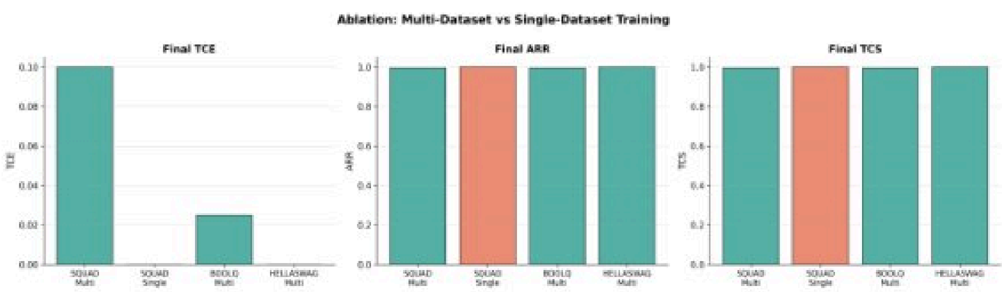

Figure 10: Comparison of multi-dataset versus single-dataset training. Multi-dataset training (solid lines) shows slightly better stability and generalization compared to single-dataset training (dashed lines), particularly evident in SQuAD where late-training degradation is less severe.

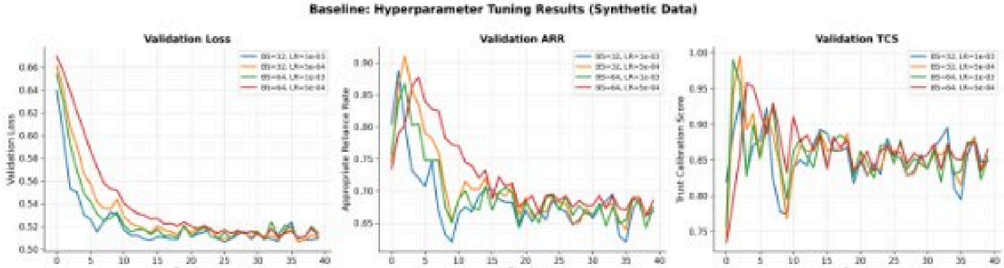

Figure 11: Hyperparameter tuning results showing final TCE across different learning rates (0.0001, 0.0005, 0.001) and hidden dimensions (32, 64, 128). Learning rate 0.0005 with hidden dimension 64 achieves the best trade-off between convergence speed and final performance across all datasets.

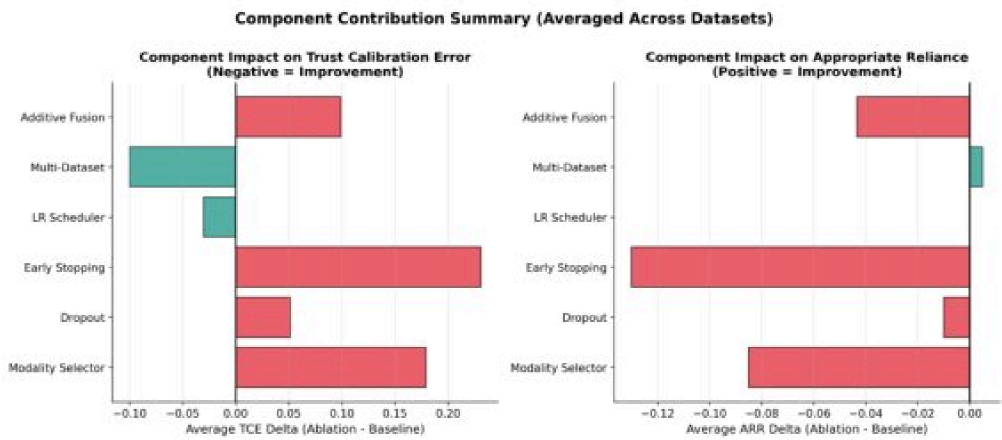

Figure 12: Component contribution ranking showing the relative importance of each component for both TCE reduction and ARR improvement. Early stopping consistently ranks highest, followed by the modality selector. Multi-dataset training shows beneficial effects particularly for TCE reduction.